Jason Cabrejos

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Robotics and embedded engineer | MS CS, BS ME | Jetson, ROS 2, GPS/RTK, machine vision

Experience

Burlington Northern Santa Fe (BNSF Railway) *March 2023 – Present*
Robotic Embedded SW Engineer

- Took a low-cost real-time inventory tracking system from prototype to production at under $10K per unit, merging IoT, networking, Power over Ethernet (PoE), and edge computing on embedded Linux (Nvidia Jetson)
- Achieved 92% container detection accuracy at production scale by combining GPS, RTK, and vision/ML models, saving $5M per year
- Scaled to a fleet of 50 deployed Jetson units across multiple hardware stacks nationwide, managed with Docker and Azure IoT Hub
- Built a Gazebo/Nav2 yard simulator and benchmarking harness to measure detection accuracy offline

Automation and Intelligent Systems Lab, UTA Research Institute *Oct 2022 – Feb 2023*
Researcher

- Developed Baxter robotic arm scripts using ROS and OpenCV for object recognition and retrieval in community-engagement projects
- Overhauled sensor stability across multiple platforms, reducing vibration-induced errors in industrial settings
- Proficient in Rapid Prototyping techniques (3d Printing, lathe, CNC and Manual Milling operations) to reduce development time

Crown Castle *January 2022 – August 2022*
Robotic Development Intern

- Reduced crew size for a fiber-optic deployment by 30% by automating industrial equipment with ROS, Python, and Ubuntu Linux
- Bridged an industrial vehicle's CAN bus to an embedded Nvidia Jetson for sensor feedback and actuator control, enabling automated curb-following and cable drift correction
- Conducted research on industrial IP-rated components ensuring its compatibility with preexisting systems for wide deployment

Laboratory of Advanced Materials Manufacturing and Analysis *June 2021 – January 2022*
Researcher

- Designed an Automated Fiber Placement (AFP) tool for a Kuka robotic arm for experimental composite layups
- Built slicer software to drive the AFP tool and robot arm together, with the Kuka ecosystem and PLCs, over EtherNet/IP and Python
- Oversaw the construction of a work cell ensuring it complied with all safety requirements and met the project's specifications

Leadership & Awards

SHPE Dallas/Fort Worth *June 2026 – Present*
Professional Development Director, Executive Board Officer

- Own the chapter's entire professional development pillar: strategy, budget, and delivery across three programs (PDS, TDS, GLEDA)
- Launched TDS (Technical Development Series) for 2026–2027, new certification-focused programming (PMP, PE, cloud, Six Sigma)

Hackathons *Various Times*
Finalist Placements

- Pitched a camera system to improve child safety without distracting the driver, 1st place
- Presented a Machine Learning model and business plan to predict potential failures on the railway using big data, 1st place
- Trained a mock interviewer LLM using Microsoft Copilot Studio and Power Automate to help practice and get feedback, 2nd place
- Trained a **Tensorflow Computer Vision** model to automate the recognition and sorting of waste using a robot, 3rd place
- Agricultural IoT platform to determine when to water/harvest using Computer Vision, Azure, and a Raspberry Pi network, 3rd place

Rover Team *August 2017 – August 2021*
President, Arm Leader, Member

- Led a team of 30 to the University Rover Challenge, traversed terrain, retrieved objects, and searched for life; placed 15th of 88 teams
- Managed a team of 5 through the design, manufacturing, and assembly of a 5 DOF Robotic Arm over the course of an academic year
- Utilized **Solidworks** and **Ansys** for modeling and Finite Element Analysis (FEA) Simulations to reach a factor of safety of 2.4
- Developed a feedback-control (PID) joint-space controller in AutoLev, simulated in MATLAB and run on an Arduino, accurate to ±0.4 in

NASA L'Space Academy *May 2019 – November 2019*
Project Manager

- Managed 2 projects: a submarine to traverse Saturn's moon Titan and improving robotic systems on the International Space Station
- Coordinated with 6 students from across the nation to achieve deadlines, share knowledge, and foster a culture of innovation
- Gained insights into NASA's approach to teamwork, workflow, planning, and budgeting in dynamic engineering setting

Education

University of Texas at Arlington *August 2025*

- Master of Science in Computer Science
  - Focus on Intelligent Systems and Computer Networks
- Bachelor of Science in Mechanical Engineering
  - Minor in Computer Science

Qualifications

- Programming Experience: Python, C++, ROS/ROS 2, Gazebo, Nav2, OpenCV, MATLAB, C#, LabVIEW
- Tools and Technologies: Nvidia Jetson, Linux, GPS/RTK, CAN bus, Git, Docker, RabbitMQ, TensorFlow, Roboflow, MongoDB, EtherNet/IP, PLCs, Azure, AWS, IoT
- Domains: Robotics, mechatronics, sensor/actuator integration, machine vision, GPS/RTK positioning, simulation
- Modeling Experience: Solidworks (Certified Associate, CSWA, issued 2016), Solid Edge